

Cancer-Specific Test Requirements

Minimum diagnostic dataset required before OncoDoseRx can generate a reliable regimen prediction.

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GASTROINTESTINAL

Colorectal Cancer

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Histopathology / Biopsy Report	Confirms tumour type, grade, differentiation, LVI, perineural invasion
 CECT Chest + Abdomen + Pelvis	Essential for staging — liver mets, lymph nodes, lung spread
 CEA (Carcinoembryonic Antigen)	Baseline tumour marker; important for treatment response monitoring
 MMR / MSI Testing (MLH1, MSH2, MSH6, PMS2)	Critical for prognosis and immunotherapy eligibility
 KRAS / NRAS / BRAF Molecular Panel	Determines eligibility for anti-EGFR therapy (cetuximab/panitumumab)
 Colonoscopy Report	Confirms tumour location, size, obstruction
 Surgical Histopathology (if surgery done)	Provides pTNM stage, node count, margin status

Required fields:

Cancer type · Stage

Important:

Grade · KRAS/NRAS · BRAF · MMR · MSI · CEA · T stage · N stage

Minimum to predict:

Cancer type

KRAS/NRAS

KRAS/NRAS status is essential for Stage IV CRC treatment selection — anti-EGFR therapy is contraindicated in RAS-mutated tumours.

MMR

MMR/MSI status affects prognosis and immunotherapy (pembrolizumab) eligibility in CRC.

STAGE

Without staging (CECT), it is impossible to determine if this is curative or palliative intent.

Rectal Cancer MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Histopathology / Biopsy Report	Confirms rectal adenocarcinoma, grade, LVI
 MRI Rectum (high-resolution)	Local staging: T-stage, mesorectal fascia, CRM involvement
 CECT Chest + Abdomen	Distant metastasis staging
 MMR / MSI Testing	Immunotherapy eligibility; dMMR rectal cancer may avoid surgery
 KRAS / NRAS / BRAF Panel	Required for metastatic disease treatment planning
 CEA	Baseline marker; elevated CEA affects prognosis

Required fields: Cancer type · Stage Important: Grade · KRAS/NRAS · BRAF · MMR · MSI · CEA · T stage · N stage Minimum to predict: Cancer type

KRAS/NRAS Molecular testing required if considering anti-EGFR therapy.

STAGE MRI rectum is the gold standard for local T/N staging in rectal cancer.

Gastric Cancer MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 OGD / Gastroscopy with Biopsy	Confirms location, histological subtype, Lauren classification
 CECT Chest + Abdomen + Pelvis	Staging, liver mets, peritoneal spread
 HER2 Testing (IHC / FISH)	HER2+ gastric cancer → trastuzumab (ToGA trial); now + nivolumab
 PD-L1 CPS Score	CPS ≥5 → nivolumab added to FOLFOX (CheckMate 649)
 MMR / MSI Testing	dMMR → pembrolizumab option

Required fields: Cancer type · Stage Important: HER2 · PD-L1 · MMR · MSI Minimum to predict: Cancer type

HER2 HER2 testing is mandatory in all advanced gastric cancer per NCCN/ESMO guidelines.

PD-L1 PD-L1 CPS determines nivolumab + FOLFOX eligibility (CheckMate 649).

Pancreatic Cancer MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 EUS-guided Biopsy / CT-guided Biopsy	Confirms PDAC vs other pancreatic tumours
 CECT Pancreas Protocol (multiphase)	Resectability assessment — SMA, SMV, portal vein involvement
 CA 19-9 (tumour marker)	Baseline marker; used for treatment monitoring
 BRCA1/BRCA2 / PALB2 Germline Testing	BRCA-mutated pancreatic cancer responds to olaparib maintenance (POLO trial)
 MSI / MMR Testing	dMMR pancreatic cancer eligible for pembrolizumab

Required fields: Cancer type · Stage **Important:** Grade · CA 19-9 · BRCA **Minimum to predict:** Cancer type

STAGE CT pancreas protocol is the first-line investigation — determines if the tumour is resectable, borderline, or metastatic.

BRCA BRCA-mutated PDAC can receive olaparib maintenance after platinum-based chemotherapy.

Hepatocellular Carcinoma (HCC)

TEST / REPORT	WHY IT IS NEEDED
 CT/MRI with contrast (LI-RADS criteria)	HCC can be diagnosed radiologically — biopsy not always needed
 AFP (Alpha-Fetoprotein)	Baseline tumour marker; very high AFP suggests HCC
 Liver Function Tests + Child-Pugh Score	Determines eligibility for systemic therapy vs TACE vs ablation
 HBV / HCV Serology	Aetiology affects prognosis; antiviral therapy required if HBV active

Required fields: Cancer type · Stage **Important:** AFP · Grade **Minimum to predict:** Cancer type

STAGE BCLC staging (Barcelona Clinic Liver Cancer) requires imaging and liver function — cannot stage without both.

Breast Cancer		
TEST / REPORT		WHY IT IS NEEDED
	Core Needle Biopsy / Histopathology PRIMARY	Tumour type, grade, Ki67, lymphovascular invasion
	Receptor Testing (ER, PR, HER2 IHC/FISH) PRIMARY	Defines breast cancer subtype — directly determines chemotherapy regimen
	CT Chest + Abdomen + Pelvis (or staging MRI) PRIMARY	Distant staging for Stage II–IV
	Genomic Risk Score (Oncotype DX / MammaPrint) CONDITIONAL	Recommended for ER+/HER2- early-stage cancers to predict recurrence risk and determine if chemotherapy can be safely avoided
	BRCA1/BRCA2 Germline Testing CONDITIONAL	Required for young patients, strong family histories of breast/ovarian cancer, or to evaluate eligibility for PARP inhibitors
	Sentinel Node Biopsy / Axillary Evaluation CONDITIONAL	Performed during or prior to surgery if imaging does not clearly define node involvement
Required fields: Cancer type · Stage Important: HER2 · ER · PR · HR · Grade · Ki-67 · BRCA Minimum to predict: Cancer type · HER2		
HER2 HER2 status is the single most important factor — HER2+ requires trastuzumab-based regimen (AC-THP), HER2- does not.		
ER ER/PR status determines whether hormonal therapy is added to chemotherapy.		
STAGE Without staging scans, it is unknown whether the disease is localised or metastatic.		

Triple-Negative Breast Cancer (TNBC)

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Biopsy / Histopathology	Grade, Ki67, LVI confirmation
 ER / PR / HER2 Confirmation	Must confirm triple-negative status (IHC + FISH)
 PD-L1 Testing (CPS score)	Determines pembrolizumab eligibility (KEYNOTE-522)
 BRCA1/BRCA2 Testing	BRCA-mutated TNBC eligible for olaparib (OlympiAD trial)
 CECT Chest + Abdomen	Staging for distant metastasis

Required fields: Cancer type · Stage

Important: PD-L1 · BRCA · Grade

Minimum to predict: Cancer type

PD-L1

PD-L1 CPS score determines whether pembrolizumab should be added (KEYNOTE-522 / KEYNOTE-355).

BRCA

BRCA mutation enables olaparib use in metastatic TNBC.

Non-Small Cell Lung Cancer (NSCLC)

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Bronchoscopy Biopsy / CT-guided Biopsy	Histological confirmation: adenocarcinoma vs squamous vs NOS
 PET-CT or CECT Chest + Abdomen + Pelvis	Accurate mediastinal and distant staging
 Molecular Panel (EGFR, ALK, ROS1, BRAF, MET, RET, KRAS G12C)	Mandatory for all non-squamous NSCLC — determines TKI eligibility
 PD-L1 TPS Score	≥50% → pembrolizumab monotherapy; 1–49% → combination regimen
 MRI Brain	Brain metastasis staging, especially for Stage IV

Required fields:

Cancer type · Stage

Important:

EGFR · ALK · PD-L1 · ROS1 · BRAF · MET

Minimum to predict:

Cancer type · Stage

EGFR

EGFR mutation is the most actionable biomarker — exon 19 del / L858R → osimertinib (FLAURA). Without EGFR status, a TKI cannot be safely recommended.

PD-L1

PD-L1 TPS determines pembrolizumab eligibility. Without this, immunotherapy planning is incomplete.

ALK

ALK rearrangement → alectinib/brigatinib (highly effective). Must be tested in all non-squamous NSCLC.

Lung Adenocarcinoma (NSCLC)

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Biopsy / Histopathology	Adenocarcinoma confirmation, TTF-1 staining
 Comprehensive Molecular Panel	EGFR, ALK, ROS1, BRAF, MET, KRAS — all mandatory in adenocarcinoma
 PD-L1 TPS	Immunotherapy eligibility
 PET-CT / CECT	Full staging

Required fields:

Cancer type · Stage

Important:

EGFR · ALK · PD-L1 · BRAF · ROS1

Minimum to predict:

Cancer type · Stage

EGFR

EGFR mutation testing is mandatory in all lung adenocarcinomas per NCCN guidelines.

Small Cell Lung Cancer (SCLC)

TEST / REPORT	WHY IT IS NEEDED
 Biopsy (bronchoscopy or CT-guided)	SCLC confirmation — synaptophysin, chromogranin, CD56 IHC
 CECT Chest + Abdomen	Limited vs extensive stage determination
 MRI Brain	Brain metastasis — very common in SCLC
 PET Scan (if limited stage)	Precise limited-stage definition for radical CRT planning

Required fields: Cancer type · Stage **Important:** PD-L1 **Minimum to predict:** Cancer type · Stage

STAGE Limited vs extensive stage distinction is critical — completely different treatment approach (CRT + PCI vs systemic + atezolizumab).

GENITOURINARY

Prostate Cancer

TEST / REPORT	WHY IT IS NEEDED
 Prostate Biopsy (TRUS or MRI-guided)	Gleason score / ISUP grade group — defines risk category
 Multiparametric MRI (mpMRI) Prostate	Local staging, extracapsular extension, seminal vesicle involvement
 PSA + PSA Kinetics	Absolute PSA, velocity, doubling time — critical for risk stratification
 Bone Scan / PSMA PET-CT	Bone and lymph node metastasis staging
 Germline BRCA / HRD Testing	BRCA2 mutation → olaparib eligibility (PROfound trial)

Required fields: Cancer type **Important:** Grade · PSA · Stage **Minimum to predict:** Cancer type

GRADE Gleason/ISUP grade group is essential — Grade Group 1 is managed very differently from Grade Group 5.

STAGE Without staging (PSMA PET-CT or bone scan), localised vs metastatic disease cannot be determined.

Bladder Cancer

TEST / REPORT	WHY IT IS NEEDED
 TURBT (Transurethral Resection) Specimen	T-stage, grade, LVI, variant histology, muscle invasion status
 CECT Abdomen + Pelvis	Lymph node staging, upper tract evaluation
 PD-L1 Testing	Avelumab maintenance eligibility after platinum chemotherapy

Required fields: Cancer type · Stage **Important:** Grade · PD-L1 **Minimum to predict:** Cancer type

STAGE Muscle invasion (T2+) vs non-muscle invasive disease is the critical decision point.

Ovarian Cancer

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Histopathology (surgery or biopsy)	Subtype: HGSC vs LGSC vs clear cell vs mucinous — changes treatment
 CECT Chest + Abdomen + Pelvis	Disease extent for upfront vs interval debulking decision
 CA-125 (tumour marker)	Baseline; used for response monitoring and recurrence detection
 BRCA1/BRCA2 Germline + Somatic Testing	BRCA mutation → PARP inhibitor maintenance (olaparib, niraparib)
 HRD Testing (Homologous Recombination Deficiency)	HRD-positive tumours respond better to PARP inhibitors even without BRCA mutation

Required fields: Cancer type · Stage Important: BRCA · Grade · CA-125 Minimum to predict: Cancer type

BRCA BRCA status is mandatory — determines PARP inhibitor maintenance eligibility (standard of care in 2024).

Cervical Cancer

TEST / REPORT	WHY IT IS NEEDED
 Cervical Biopsy / Punch Biopsy	SCC vs adenocarcinoma — different prognosis and management
 MRI Pelvis	Local T-staging, parametrial extension, uterine involvement
 CECT Chest + Abdomen	Lymph node and distant metastasis staging
 PD-L1 CPS Score	PD-L1+ → pembrolizumab + carboplatin + paclitaxel ± bev (KEYNOTE-826)

Required fields: Cancer type · Stage Important: PD-L1 · Grade Minimum to predict: Cancer type · Stage

PD-L1 PD-L1 CPS determines pembrolizumab eligibility in metastatic/recurrent cervical cancer.

STAGE FIGO staging requires MRI pelvis — clinical staging alone is insufficient.

HAEMATOLOGICAL

Non-Hodgkin Lymphoma

TEST / REPORT	WHY IT IS NEEDED
 Excision Biopsy / Core Biopsy	Lymphoma subtype classification — DLBCL, follicular, mantle cell, etc.
 Immunophenotyping (IHC / Flow Cytometry)	CD20, BCL2, BCL6, MYC — defines subtype and treatment
 PET-CT (FDG)	Ann Arbor staging; baseline for response assessment
 Bone Marrow Biopsy	Stage IV determination; treatment planning
 LDH + IPI Score	International Prognostic Index — determines prognosis and intensity

Required fields:

Cancer type · Stage

Important:

CD20 · Grade

Minimum to predict:

Cancer type · Stage

CD20

CD20 status determines whether rituximab can be added (R-CHOP vs CHOP).

STAGE

PET-CT staging is mandatory before starting R-CHOP.

Hodgkin Lymphoma

TEST / REPORT	WHY IT IS NEEDED
 Excision Biopsy	Classic HL (RS cells) vs nodular lymphocyte-predominant HL
 PET-CT (FDG)	Ann Arbor staging; interim PET after Cycle 2 guides treatment
 Bone Marrow Biopsy (if PET non-avid)	Stage IV determination in non-FDG-avid disease

Required fields:

Cancer type · Stage

Important:

Grade

Minimum to predict:

Cancer type · Stage

STAGE

Ann Arbor stage determines number of cycles (4 vs 6) and whether radiotherapy is added.

Leukemia (AML)

MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Bone Marrow Biopsy + Aspirate	Blast percentage, morphology, cytochemistry
 Cytogenetics (Karyotype)	Favourable / intermediate / adverse — defines risk and consolidation
 Molecular Panel (FLT3-ITD, NPM1, IDH1/2, CEBPA)	FLT3-ITD → midostaurin; IDH1 → ivosidenib; IDH2 → enasidenib
 CBC + Peripheral Blood Film	Blast count, pancytopenia degree

Required fields:

Cancer type

Important:

Grade · FLT3 · NPM1 · IDH

Minimum to predict:

Cancer type

FLT3

FLT3-ITD mutation determines midostaurin addition to 7+3 induction (RATIFY trial).

Multiple Myeloma MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 Bone Marrow Biopsy (trephine + aspirate)	Plasma cell percentage, cytogenetics
 SPEP / UPEP + Free Light Chains	M-protein quantification — diagnosis and response monitoring
 FISH Cytogenetics (del17p, t(4;14), t(14;16))	High-risk cytogenetics affect VRd intensity and ASCT timing
 PET-CT or Whole Body Low-dose CT	Bone disease, plasmacytomas
 LDH + Beta-2 Microglobulin + Albumin (ISS staging)	ISS/R-ISS staging determines prognosis

Required fields: Cancer type **Important:** Stage · Grade **Minimum to predict:** Cancer type

STAGE ISS/R-ISS staging uses B2M + albumin + LDH + FISH — determines VRd intensity.

CENTRAL NERVOUS SYSTEM

Brain Cancer MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 MRI Brain with Contrast	Essential for diagnosis, grading, and surgical planning — defines tumor type, size, and edema
 Surgical Resection / Biopsy Specimen	Histopathology confirms tumor type (GBM, LGG, oligodendroglioma, etc.) and WHO grade
 MGMT Promoter Methylation Testing	Critical for GBM — predicts temozolomide response and prognosis
 Molecular Panel (IDH1/2, 1p/19q, TERT, ATRX)	WHO 2021 classification — IDH status, 1p/19q co-deletion, TERT promoter mutation
 Neurological Assessment + KPS/ECOG	Baseline functional status — affects treatment intensity and clinical trial eligibility

Required fields: Cancer type · Stage **Important:** Grade · MGMT · IDH · WHO CNS grade · Extent of resection **Minimum to predict:** Cancer type

MGMT MGMT promoter methylation is the most important prognostic biomarker in GBM — determines temozolomide benefit.

STAGE MRI brain is essential for staging — defines resectability, eloquent cortex involvement, and need for adjuvant therapy.

IDH IDH mutation status reclassifies gliomas per WHO 2021 — IDH-mutant has better prognosis and different treatment.

Brain Tumor MOLECULAR PANEL REQUIRED

TEST / REPORT	WHY IT IS NEEDED
 MRI Brain with Contrast	Essential for diagnosis, grading, and surgical planning
 Surgical Resection / Biopsy Specimen	Histopathology confirms tumor type and WHO grade
 MGMT Promoter Methylation Testing	Critical for GBM — predicts temozolomide response
 Molecular Panel (IDH1/2, 1p/19q, TERT, ATRX)	WHO 2021 classification markers
 Neurological Assessment + KPS/ECOG	Baseline functional status

Required fields: Cancer type · Stage Important: Grade · MGMT · IDH · WHO CNS grade · Extent of resection Minimum to predict: Cancer type

- MGMT** MGMT promoter methylation is the most important prognostic biomarker in GBM.
- STAGE** MRI brain is essential for staging and treatment planning.
- IDH** IDH mutation status reclassifies gliomas per WHO 2021.

FALLBACK — ANY CANCER NOT LISTED ABOVE

Default pathway

TEST / REPORT	WHY IT IS NEEDED
 Histopathology / Biopsy Report	Confirms cancer type, grade, histological subtype
 CECT / MRI (staging)	Disease extent, lymph node and distant staging

Required fields: Cancer type · Stage Important: Grade · Histology Minimum to predict: Cancer type